Constructions — Practice — Answer Key

You need a compass, a straightedge and a pencil. A worked example comes before each type.

Name: _____ Date: _____

Type 1 Tools & Words

Watch Me — Worked Example

In a construction, may you use the inch marks on a ruler?
No — a straightedge only draws straight lines. Lengths are copied with the compass. **Answer: no**

1. Which tool copies a length?

compass

2. Which tool draws straight lines?

straightedge

3. A bisector cuts a segment or an angle into two _____ parts.

equal

4. The perpendicular bisector meets the segment at its _____, at a _____ angle.

midpoint, right ($90°$)

Type 2 Choose the Compass Width

Watch Me — Worked Example

To bisect a 10 cm segment, how wide must the compass be?
More than half of 10 cm, so more than 5 cm (for example 6 cm). **Answer: more than 5 cm**

5. A segment is 12 cm. Which width works: 5 cm or 7 cm?

7 cm

6. A segment is 9 cm. What is the smallest whole-cm width that works?

5 cm

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7. Why must both arcs use the **same** width?

8. On a number line A is at 2 and B is at 14. Where does the perpendicular bisector cross?

so P and Q are equally far from A and B

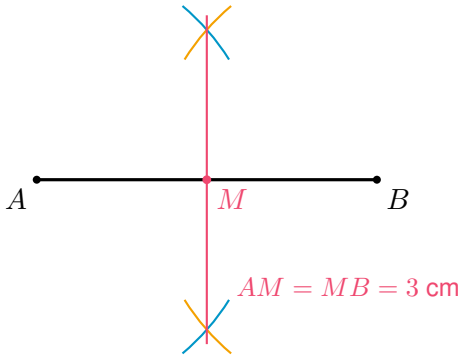
at 8

Type 3 Construct It

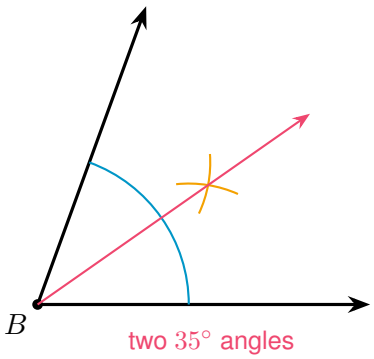
Watch Me — Worked Example

Leave every arc on the page — the arcs are your "show your work." Check with a ruler only at the very end.

9. Bisect $\overline{AB}$ (it is 6 cm). Label the midpoint M.



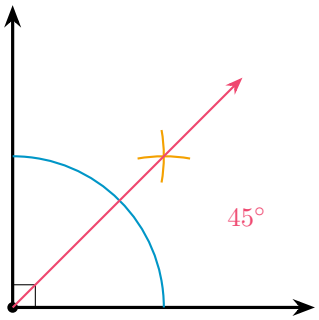
10. Bisect $\angle B$ (it is $70°$).



11. Copy $\overline{AB}$ onto the ray, starting at C. Label the new endpoint D.



12. HARD Bisect the right angle to make a $45°$ angle.



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Type 4 Use What a Bisector Does

Watch Me — Worked Example

M is the midpoint of $\overline{AB}$. $AM = 2x + 3$ and $MB = 4x - 5$. Find AB.
A midpoint makes equal halves: $2x + 3 = 4x - 5$, so $x = 4$. Each half is 11, so $AB = 22$. **Answer: 22**

13. An $84°$ angle is bisected. How big is each half?

$42°$

14. M is the midpoint of $\overline{AB}$. $AM = 3x + 1$, $MB = 5x - 7$. Find AB.

$AB = 26$

15. $\overrightarrow{BZ}$ bisects $\angle ABC$. $m\angle ABZ = 2x + 10$, $m\angle ZBC = 4x - 6$. Find $m\angle ABC$.

$x = 8,\ 52°$

16. Why is the crossing point P the same distance from A and from B?

both arcs have the same radius

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